
Simulation of Flow and Transport in the Partially Molten Mantle of the Earth

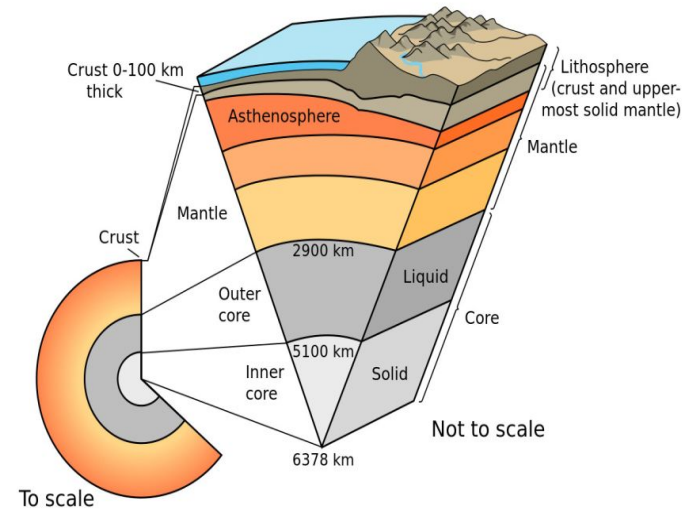
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Partial melting in the Asthenosphere is a crucial geophysical process in the lithospheric evolution and plate tectonics.



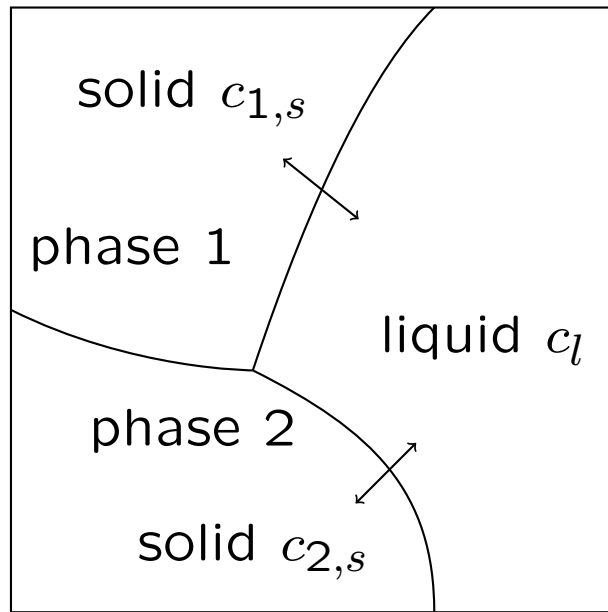
(National Geographic)

Goal. Provide an **accurate numerical approximation** of the time-dependent dynamics of a partially molten mantle.

Objective. Provide a **locally mass conservative** numerical framework for

- The mechanics of a deformable solid matrix that is possibly partially molten.
- The transport phenomena of chemical species and energy in the presence of porous media.

Elementary Representative Volume.



Mixture Theory

Both liquid (l) and solid (s) phases exist at each point of the domain Ω .

Define the **mixture variable** (phase averaged) as:

$$\begin{aligned}\bar{\Psi} &= \phi \Psi_l + (1 - \phi) \Psi_s \\ &= \Psi_s + \phi \Psi_r, \\ \Psi_r &= \Psi_l - \Psi_s.\end{aligned}$$

Extended Boussinesq Approximation. Density difference between phases are neglected expect in the **body force term**.

Notation.

c Mass concentration. p Pressure.
 ρ Density. μ Viscosity.
 h Specific enthalpy. $\mathbf{v}$ Velocity.

T Temperature.
 $\mathbf{g}$ Gravity.
 L Latent heat.

c_p Specific heat.
 $\mathbf{g}$ Gravity.
 α_ρ Thermal expansion.

McKenzie Equations

A two-phase formulation of a viscous mantle.

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \quad (\text{Darcy's Law})$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho} \mathbf{g} = 0, \quad (\text{Stokes Flow})$$

$$\frac{\partial \rho_l \phi}{\partial t} + \nabla \cdot (\rho_l \phi \mathbf{v}_l) = \Gamma,$$

$$\frac{\partial \rho_s(1 - \phi)}{\partial t} + \nabla \cdot (\rho_s(1 - \phi) \mathbf{v}_s) = -\Gamma,$$

Deviatoric stress of mixture

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s \phi_s \left(\frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T) - \frac{1}{3} \nabla \cdot \mathbf{v}_s \mathbf{I} \right), \quad (1)$$

Unknowns.

Four equations and six unknowns.

- Velocities $\mathbf{v}_l$ and $\mathbf{v}_s$.
- Pressure p_l and p_s .
- Porosity ϕ .
- Melting source Γ .

Three parts.

- Static mechanic equations.
- Transport of chemical species and energy.
- Phase package or prescribed melting function.

1. Static System



Pressure Potentials.

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z,$$

$$p_l - p_s = q_l - q_s = \frac{1}{1 - \phi}(q_l - q), \quad q = \bar{q}.$$

Then

$$\phi \mathbf{v}_r + \frac{k_0 \phi^{2+2\Theta}}{\mu_l} \nabla q_l = 0, \quad \text{(Darcy's Law)} \quad (2)$$

$$\mu_s \nabla \cdot (\phi \mathbf{v}_r) + \frac{\phi}{1 - \phi}(q_l - q) = 0, \quad \text{(Conservation / Compaction)} \quad (3) \quad (4)$$

$$\nabla q - \nabla \cdot \tau(\mathbf{v}_s) = -(1 - \phi) \rho_r \mathbf{g}, \quad \text{(Stokes Flow)} \quad (5)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1 - \phi}(q_l - q) = 0, \quad \text{(Compaction)} \quad (6)$$

Problems. When no melt, $\phi = 0$,

1. Darcy system degenerates when ϕ vanishes, lose bound on q_l .
2. Lose the conservation equation.
3. The condition number grows as $\phi \rightarrow 0$.

Scaled Variables

Define by Arbogast and Taicher [1]:

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q} = \phi^{1/2} q_l,$$

The Space $H_\phi(\text{div})$

$H_\phi(\text{div}; \Omega)$ is a Hilbert space

$$H_\phi(\text{div}; \Omega) = \left\{ \mathbf{v} \in (L^2(\Omega))^2 : \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \mathbf{v}) \in L^2(\Omega) \right\}.$$

equipped with the weighted inner-product

$$(\mathbf{u}, \mathbf{v})_{H_\phi(\text{div})} = (\mathbf{u}, \mathbf{v}) + \left(\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \mathbf{u}), \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \mathbf{v}) \right).$$

Moreover, $H(\text{div}; \Omega) = H_1(\text{div}; \Omega) \subset H_\phi(\text{div}; \Omega)$.

The normal trace on $\partial\Omega$ is well-defined by

$$\langle \phi^{1/2+\Theta} \mathbf{v} \cdot \boldsymbol{\nu}, \psi \rangle = \left(\mathbf{v}, \phi^{1+\Theta} \nabla (\phi^{-1/2} \psi) \right) + \left(\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \mathbf{v}), \psi \right), \quad \forall \psi \in H^1(\Omega)$$

A Dimensionless Boundary Value Problem

Find two velocity-pressure potential pairs $(\tilde{\mathbf{v}}_r, \tilde{q}_l)$ and $(\tilde{\mathbf{v}}_s, \tilde{q})$ such that

$$\tilde{\mathbf{v}}_r + \phi^{1+\Theta} \nabla(\phi^{-1/2} \tilde{q}_l) = 0, \quad (7)$$

$$\phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi} (\tilde{q}_l - \phi^{1/2} \tilde{q}) = 0, \quad (8)$$

$$\nabla \tilde{q} - \nabla \cdot \tilde{\boldsymbol{\tau}}(\tilde{\mathbf{v}}_s) = -(1-\phi) \mathbf{n}, \quad (9)$$

$$\nabla \cdot \tilde{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi} (\tilde{q}_l - \phi^{1/2} \tilde{q}) = 0, \quad (10)$$

and

$$\begin{aligned} \phi^{1+\Theta} \tilde{\mathbf{v}}_r \cdot \boldsymbol{\nu} &= g_r && \text{on } \Gamma_{u,D}, \\ \tilde{\mathbf{v}}_s &= \mathbf{g}_s && \text{on } \Gamma_{u,S}, \\ \tilde{q}_l &= \phi^{1/2} q_{l,0} && \text{on } \Gamma_{i,D}, \\ \boldsymbol{\tau} \cdot \boldsymbol{\nu} - q &= t_0 && \text{on } \Gamma_{i,S}. \end{aligned} \quad (11)$$

Characteristic Scales.

- $l_0 = \left(\frac{k_0 \mu_s}{\mu_l} \right)^{1/2}, \quad u_0 = \frac{k_0 \rho_r g}{\mu_l}.$
- $t_0 = l_0 / u_0, \quad p_0 = \rho g l_0.$

Dimensionless Variables.

- $\tilde{\mathbf{v}}_r = \frac{\tilde{\mathbf{v}}_r}{u_0}, \quad \tilde{\mathbf{v}}_s = \frac{\mathbf{v}_s}{u_0},$
- $\tilde{q} = \frac{q}{p_0}, \quad \tilde{q}_l = \frac{\tilde{q}_l}{p_0}.$

Element averaged variable:

$$\phi_E = \frac{1}{|E|} \int_E \phi(\mathbf{x}) d\mathbf{x},$$

Scaled weak form Modified to be locally mass conservative:

$$(\boldsymbol{\tau}(\tilde{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\tilde{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1 - \phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}) \quad \forall \boldsymbol{\psi}_{s,h} \in \mathbb{V}_{S,h,0},$$

$$(\nabla \cdot \tilde{\mathbf{v}}_{s,h}, \omega_h) - \left(\frac{\phi \phi_E^{-1/2}}{1 - \phi} (\tilde{q}_{l,h} - \phi_E^{1/2} \tilde{q}_h), \omega_h \right) = 0, \quad \forall \omega_h \in \mathbb{W}_{S,h},$$

$$(\tilde{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\tilde{q}_{l,h}, \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h})) = 0, \quad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0},$$

$$(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_{r,h}), \omega_{l,h}) + \left(\frac{\phi \phi_E^{-1}}{1 - \phi} (\tilde{q}_{l,h} - \phi_E^{1/2} \tilde{q}_h), \omega_{l,h} \right) = 0, \quad \forall \omega_{l,h} \in \mathbb{W}_{D,h}.$$



2. Transport System



An advection-diffusion model

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot (\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}) = 0,$$

Phase averaged quantities:

$$\overline{\rho c_i} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j}, \quad \overline{\rho c_i \mathbf{v}} = \sum_{j \in \{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j.$$

Separate phase averaged concentration

$$\overline{c_i \mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi \mathbf{v}_l + D(1 - \phi) \mathbf{v}_s}{\phi + (1 - \phi)D} \right) \bar{c}_i.$$

The partition coefficient $D = c_{i,s}/c_{i,l}$.

Effective Velocity

$$\mathbf{v}_{\text{eff}} = \frac{\phi \mathbf{v}_l + D(1 - \phi) \mathbf{v}_s}{\phi + (1 - \phi)D}$$

Specific internal energy formulation:

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{K \nabla T} + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v} \cdot \mathbf{g}},$$

where $\overline{K} = \phi_l K_l + \phi_s K_s$ is the phase averaged thermal conductivity.

Temperature formulation:

$$\frac{\partial \overline{H}}{\partial t} + \nabla \cdot (\overline{\mathbf{v} T}) = \frac{L}{c_p} \nabla \cdot (\phi_s \mathbf{v}_s) + \kappa \nabla^2 T + \frac{\alpha_\rho}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g},$$

Specific enthalpy:

$$h_i = u_i + p_i \tilde{V}_i = u_i + P/\rho_i,$$

Bulk specific enthalpy:

$$\overline{H} = \overline{\rho h} / \rho c_p = \overline{h} / c_p$$

FVM for General Conservation Law





3. Coupling Strategy





